Name: **ANSWER KEY** _____ Date: _____

Quadratic Functions & Complex Numbers: Color By Number



Color Code:

9 Pink 4	**1** Red 16	**5** Purple −23	**10** Tan 9	**4** Light Blue −13
6 Orange 10	**2** Yellow −2	**8** Gray −3	**7** Brown 22	**3** Green −5

Name: **ANSWER KEY** _____ Date: _____

Quadratic Functions & Complex Numbers: Color By Number

Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch closely: the answers look alike!

1.

A ramp's profile is $y = -x^2 + 10x - 9$ (feet). Find the maximum height, in feet.

Vertex $x = -\dfrac{10}{2(-1)} = 5$

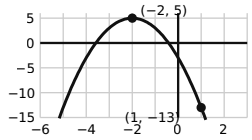
$y = -25 + 50 - 9$ (5 is the x value, not the height)

16

1. Color: **Red**

2.

A parabola has vertex $(-2, 5)$ and passes through $(1, -13)$. In $y = a(x - h)^2 + k$, find a.



$y = a(x + 2)^2 + 5$; at $(1, -13)$: $-13 = 9a + 5$

−2

2. Color: **Yellow**

3.

Solve $2x^2 + 9x - 5 = 0$ by factoring. What is the smaller solution?

$(2x - 1)(x + 5) = 0$

$x = \dfrac{1}{2}$ or $x = -5$

−5

3. Color: **Green**

4.

Write $y = 2x^2 - 12x + 5$ as $y = a(x - h)^2 + k$ by completing the square. Find k.

$y = 2(x^2 - 6x) + 5 = 2(x^2 - 6x + 9) + 5 - 2(9)$

$y = 2(x - 3)^2 - 13$ (subtract 2(9), not 9)

−13

4. Color: **Light Blue**

5.

Find the discriminant, $b^2 - 4ac$, of $3x^2 - 5x + 4 = 0$.

$a = 3, b = -5, c = 4$

$(-5)^2 - 4(3)(4) = 25 - 48$ (two complex roots)

−23

5. Color: **Purple**

6.

Solving $2x^2 - 8x + 3 = 0$ with the quadratic formula gives $x = \dfrac{4 \pm \sqrt{n}}{2}$. Find n.

$x = \dfrac{8 \pm \sqrt{40}}{4} = \dfrac{4 \pm \sqrt{10}}{2}$

10

6. Color: **Orange**

7.

Multiply $(3 - 2i)(4 + 5i)$ and write the result as $a + bi$. Find a.

$12 + 15i - 8i - 10i^2$

$i^2 = -1$, so $-10i^2 = +10$: $(12 + 10) + 7i$

22

7. Color: **Brown**

8.

Divide: $\dfrac{9 - 7i}{3 + i} = a + bi$. Find b.

Multiply by the conjugate $3 - i$:

$\dfrac{(9 - 7i)(3 - i)}{9 + 1} = \dfrac{20 - 30i}{10} = 2 - 3i$

−3

8. Color: **Gray**

9.

A ball is thrown so its height in feet after t seconds is $h = -16t^2 + 48t + 64$. How many seconds until it hits the ground?

Set $h = 0$: $-16(t^2 - 3t - 4) = 0$, so $(t - 4)(t + 1) = 0$

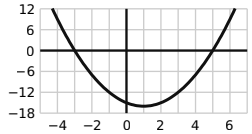
$t = 4$ or $t = -1$ (reject: time cannot be negative)

4

9. Color: **Pink**

10.

Use the graph of $y = x^2 - 2x - 15$ to find the sum of the integer solutions of $x^2 - 2x - 15 \le 0$.



$(x - 5)(x + 3) \le 0$ for $-3 \le x \le 5$; sum $= 4 + 5$

9

10. Color: **Tan**